INTERNSHIP · INDUSTRIAL AUTOMATION

Automation and supervision of a blow-room line

Sitex · Sousse · Jan – May 2024 (final-year project)

S7-1500

Siemens PLC, CPU 1512C-1 PN

GRAFCET

one per machine of the line

WinCC

HMI for monitoring and alarms

CONTEXT

A textile blow-room line (BDT, Axiflow, MPM, CVT4, DX, plus technical fans and a condenser) ran on a legacy relay system. I designed its migration to a PLC.

WHAT I DID

- Analysed each machine’s functions and safety signals, such as the emergency stop and door safety sensor.
- Wrote GRAFCET for each machine, including automatic, manual and safety modes.
- Selected the PLC and the digital and analog I/O modules.
- Programmed it in TIA Portal with structured blocks: OB1, FC1 for analog inputs, FC2 for analog outputs, plus PID control.
- Built a WinCC HMI and drew the control-cabinet wiring diagrams.

TOOLS AND METHODS

Siemens S7-1500 · TIA Portal · WinCC · GRAFCET · PID · Wiring diagrams

OUTCOME

A complete automation design, from functional analysis to PLC program and operator interface, replacing the relay logic.

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